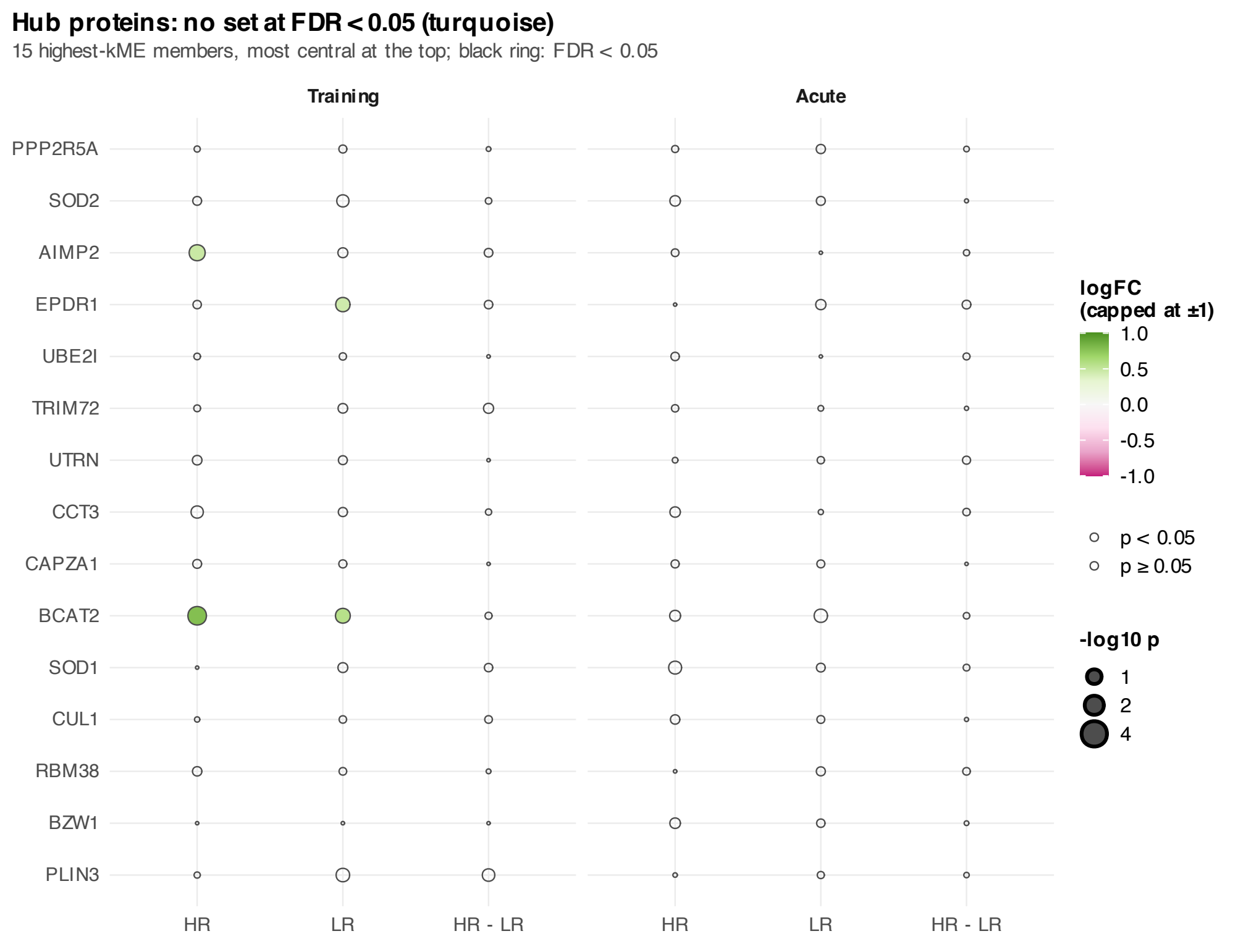
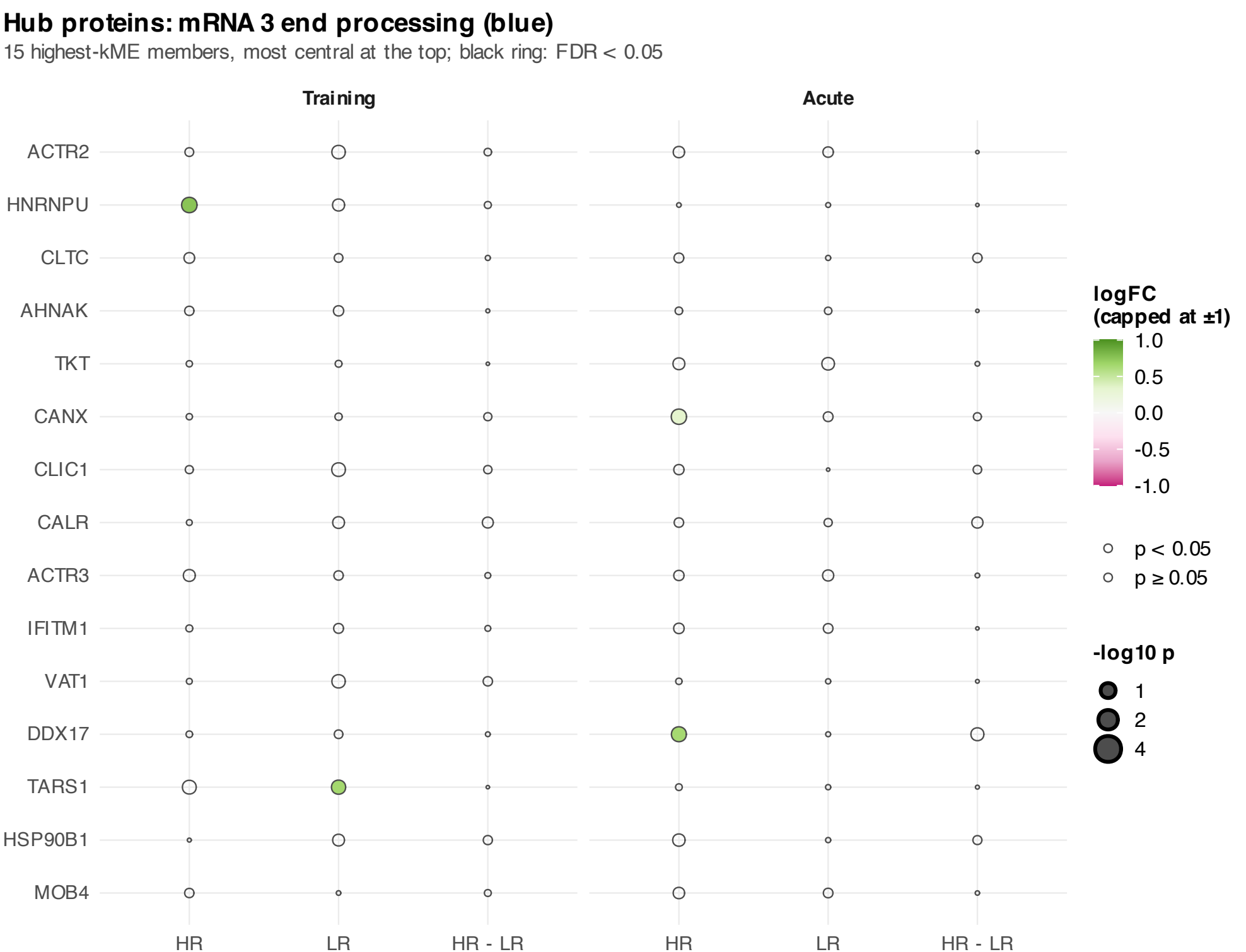
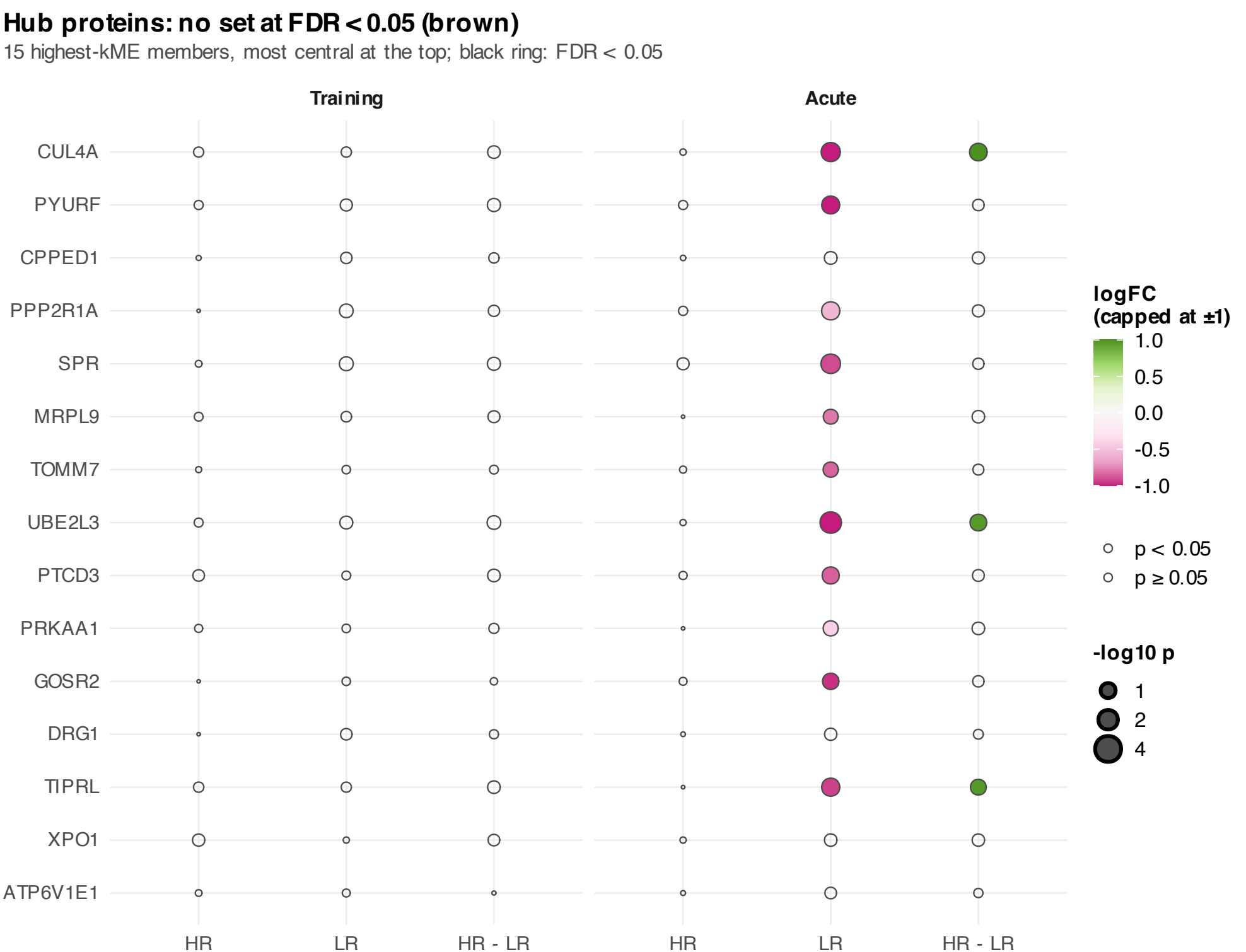
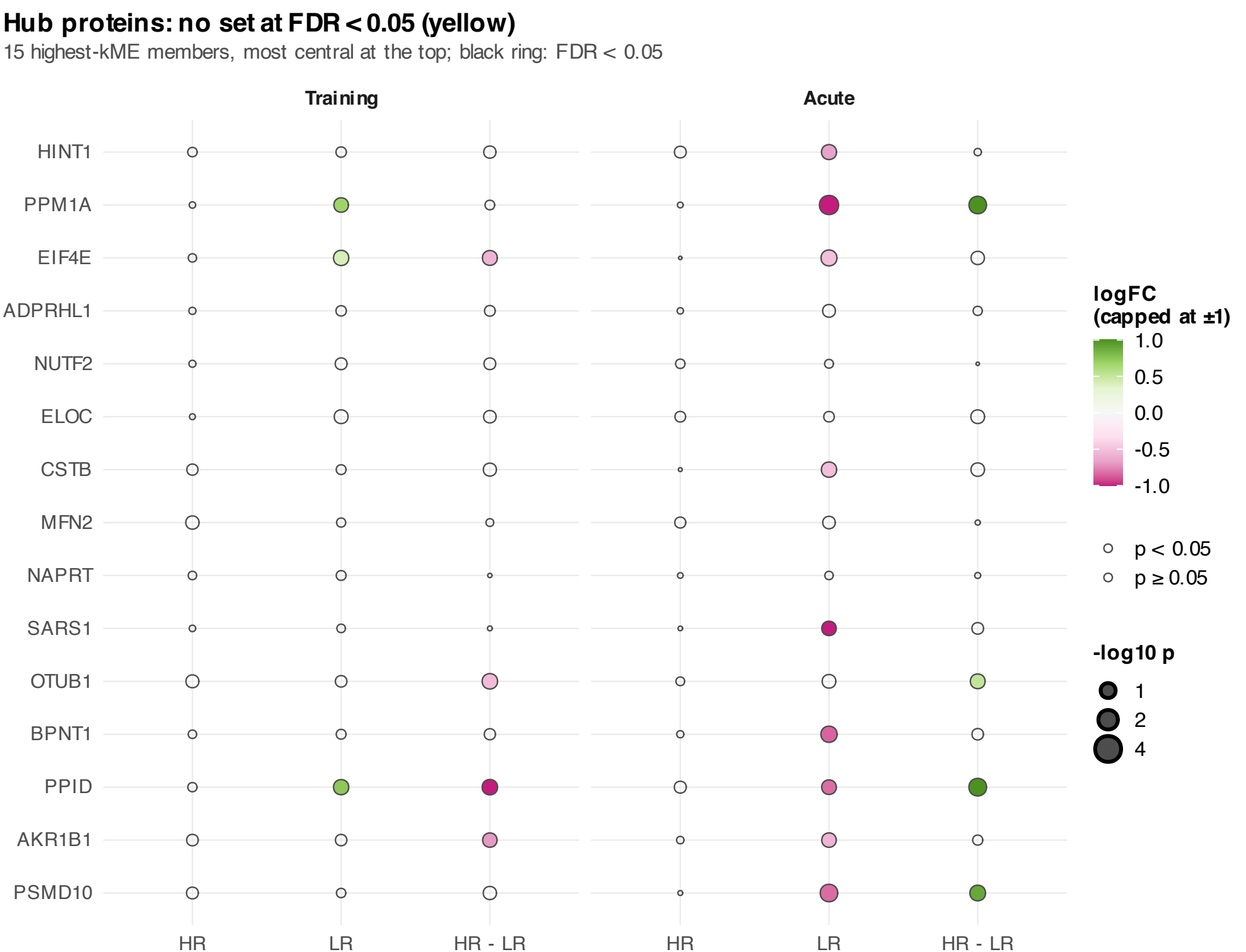
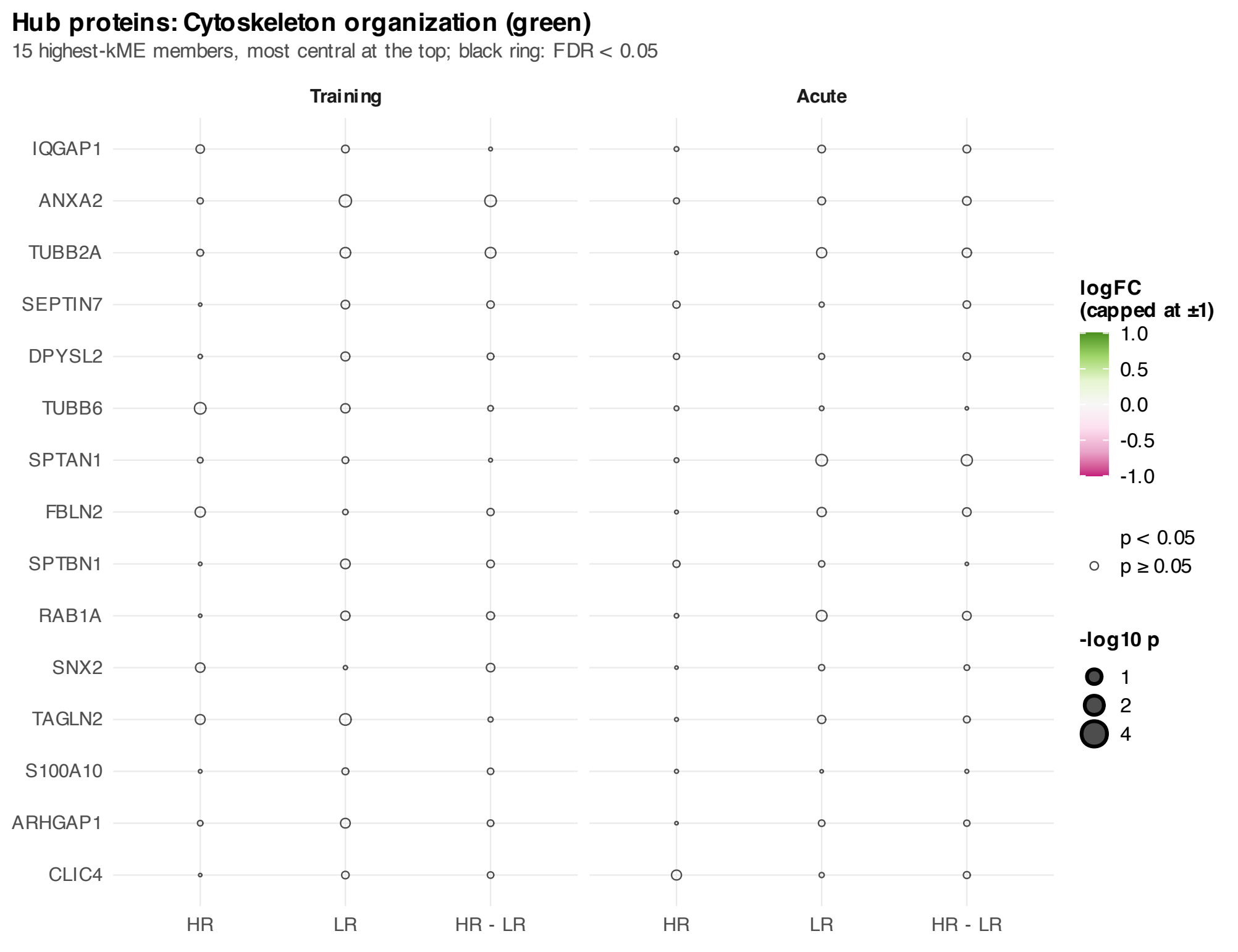
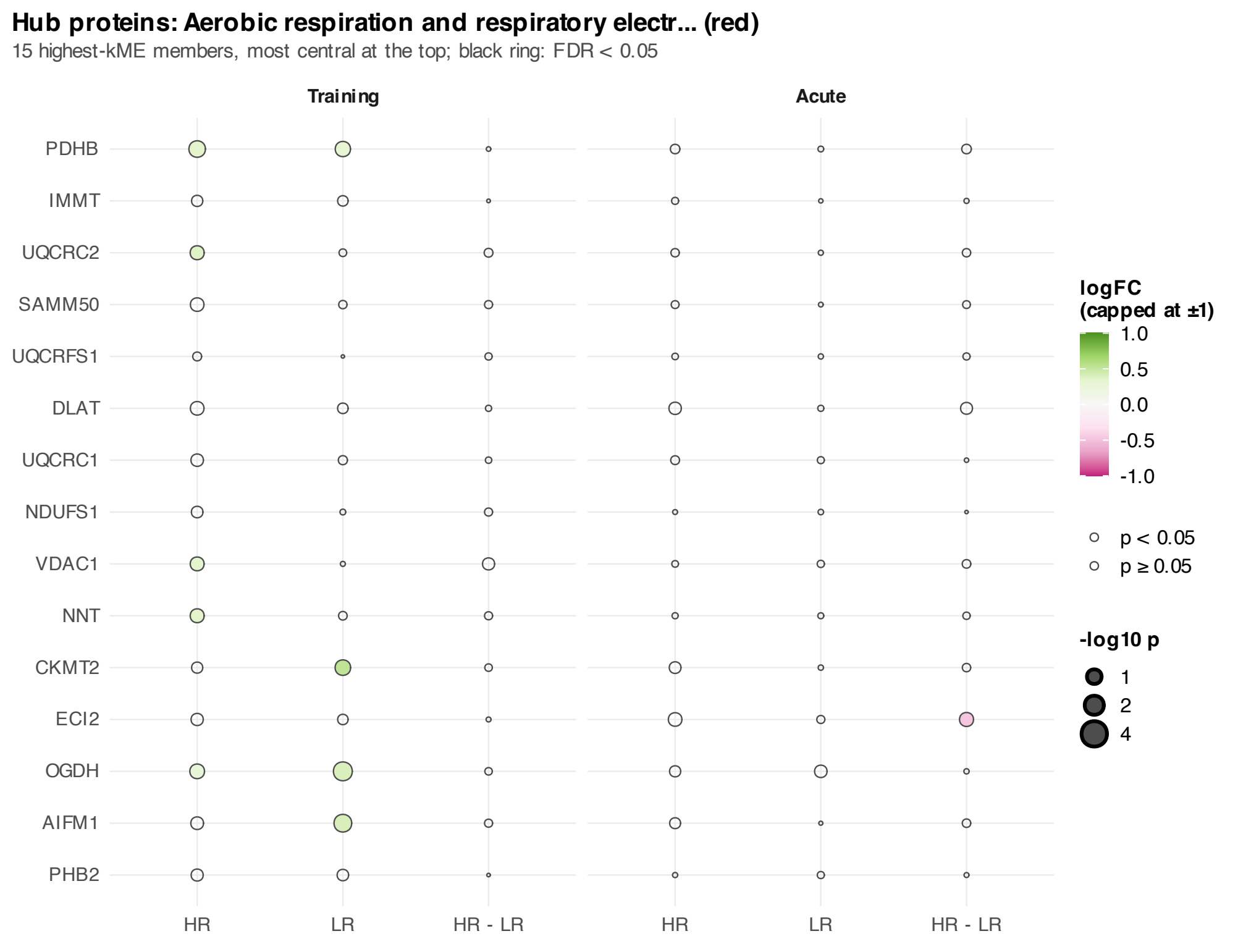

Hub proteins: no set at FDR < 0.05 (turquoise)
15 highest-kME members, most central at the top; black ring: FDR < 0.05
Training
Acute
PPP2R5A
SOD2
AIMP2
EPDR1
UBE2I
TRIM72
UTRN
CCT3
CAPZA1
BCAT2
SOD1
CUL1
RBM38
BZW1
PLIN3
HR
LR
HR - LR
HR
LR
HR - LR
logFC
(capped at ±1)
1.0
0.5
0.0
-0.5
-1.0
p < 0.05
p ≥ 0.05
-log10 p
1
2
4


Hub proteins: mRNA 3 end processing (blue)
15 highest-kME members, most central at the top; black ring: FDR < 0.05
Training
Acute
ACTR2
HNRNPU
CLTC
AHNAK
TKT
CANX
CLIC1
CALR
ACTR3
IFITM1
VAT1
DDX17
TARS1
HSP90B1
MOB4
HR
LR
HR - LR
HR
LR
HR - LR
logFC
(capped at ±1)
1.0
0.5
0.0
-0.5
-1.0
p < 0.05
p ≥ 0.05
-log10 p
1
2
4


Hub proteins: no set at FDR < 0.05 (brown)
15 highest-kME members, most central at the top; black ring: FDR < 0.05
Training
Acute
CUL4A
PYURF
CPPED1
PPP2R1A
SPR
MRPL9
TOMM7
UBE2L3
PTCD3
PRKAA1
GOSR2
DRG1
TIPRL
XPO1
ATP6V1E1
HR
LR
HR - LR
HR
LR
HR - LR
logFC
(capped at ±1)
1.0
0.5
0.0
-0.5
-1.0
p < 0.05
p ≥ 0.05
-log10 p
1
2
4

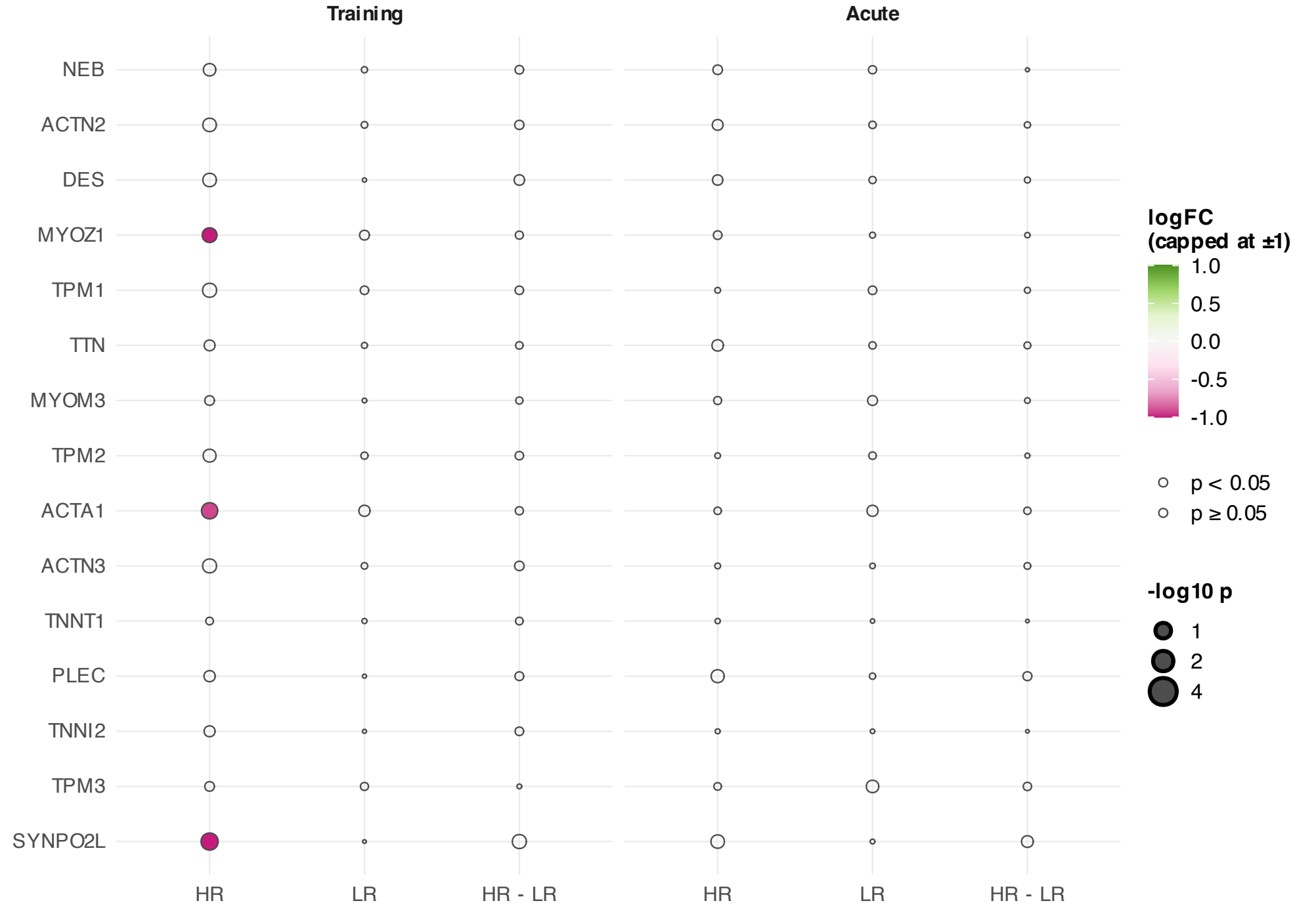

Hub proteins: no set at FDR < 0.05 (yellow)
15 highest-kME members, most central at the top; black ring: FDR < 0.05
Training
Acute
HINT1
PPM1A
EIF4E
ADPRHL1
NUTF2
ELOC
CSTB
MFN2
NAPRT
SARS1
OTUB1
BPNT1
PPID
AKR1B1
PSMD10
HR
LR
HR - LR
HR
LR
HR - LR
logFC
(capped at ±1)
1.0
0.5
0.0
-0.5
-1.0
p < 0.05
p ≥ 0.05
-log10 p
1
2
4

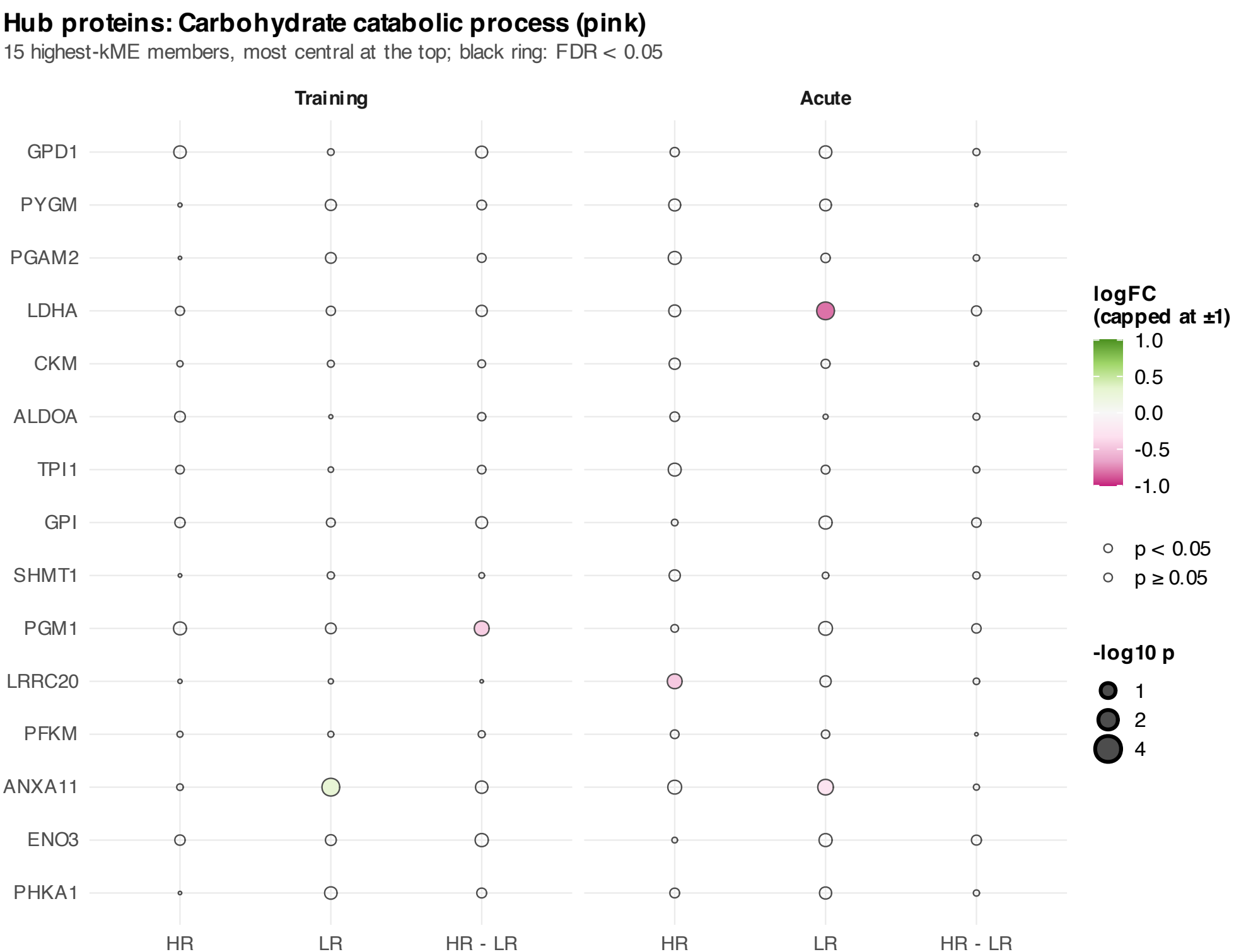

Hub proteins: Cytoskeleton organization (green)
15 highest-kME members, most central at the top; black ring: FDR < 0.05
Training
Acute
IQGAP1
ANXA2
TUBB2A
SEPTIN7
DPYSL2
TUBB6
SPTAN1
FBLN2
SPTBN1
RAB1A
SNX2
TAGLN2
S100A10
ARHGAP1
CLIC4
HR
LR
HR - LR
HR
LR
HR - LR
logFC
(capped at ±1)
1.0
0.5
0.0
-0.5
-1.0
p < 0.05
p ≥ 0.05
-log10 p
1
2
4

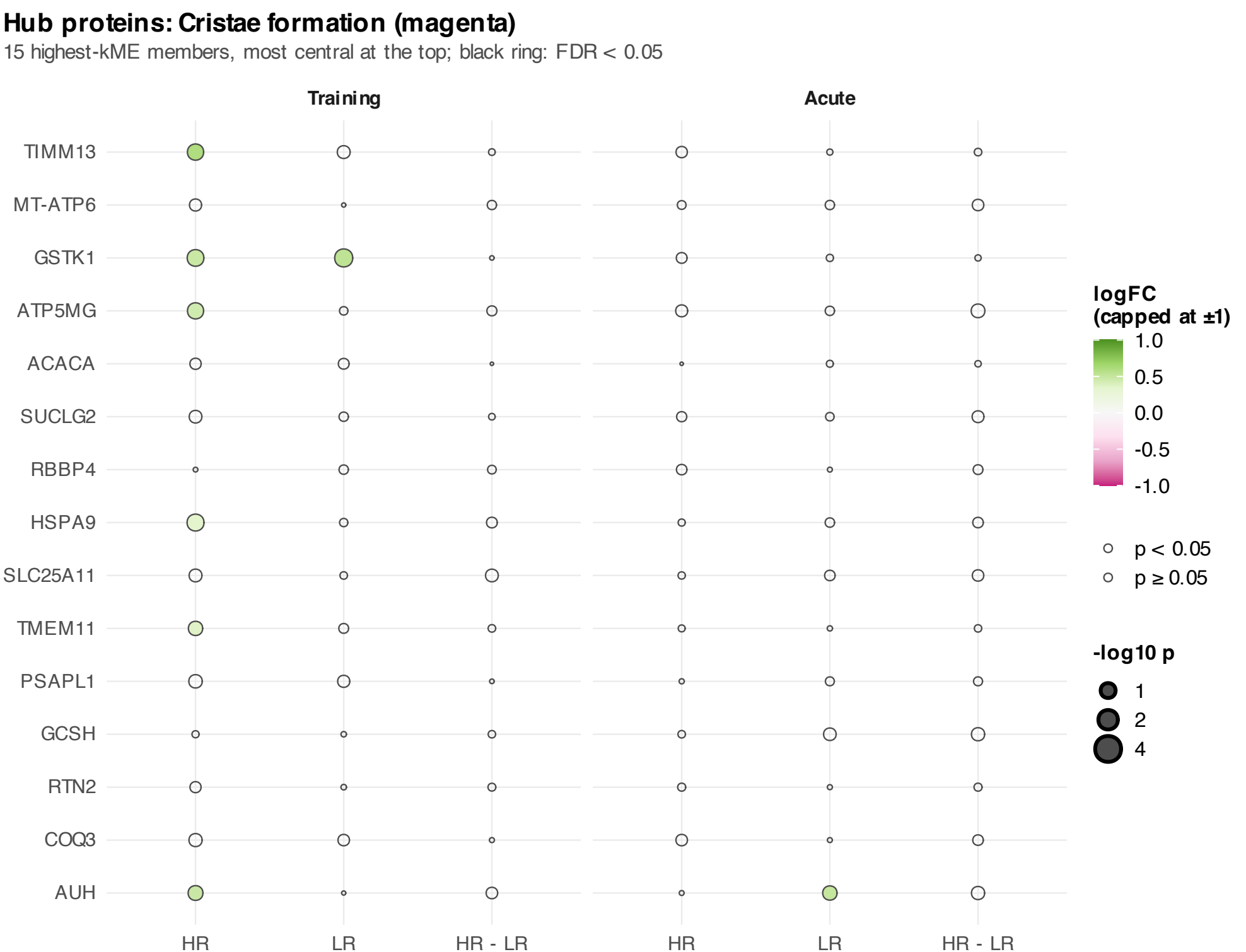

Hub proteins: Aerobic respiration and respiratory electr... (red)
15 highest-kME members, most central at the top; black ring: FDR < 0.05
Training
Acute
PDHB
IMMT
UQCRC2
SAMM50
UQCRFS1
DLAT
UQCRC1
NDUFS1
VDAC1
NNT
CKMT2
ECI2
OGDH
AIFM1
PHB2
HR
LR
HR - LR
HR
LR
HR - LR
logFC
(capped at ±1)
1.0
0.5
0.0
-0.5
-1.0
p < 0.05
p ≥ 0.05
-log10 p
1
2
4

Hub proteins: Striated muscle contraction (black)

15 highest-kME members, most central at the top; black ring: FDR < 0.05




Hub proteins: Carbohydrate catabolic process (pink)
15 highest-kME members, most central at the top; black ring: FDR < 0.05
Training
Acute
GPD1
PYGM
PGAM2
LDHA
CKM
ALDOA
TPI1
GPI
SHMT1
PGM1
LRRC20
PFKM
ANXA11
ENO3
PHKA1
HR
LR
HR - LR
HR
LR
HR - LR
logFC
(capped at ±1)
1.0
0.5
0.0
-0.5
-1.0
p < 0.05
p ≥ 0.05
-log10 p
1
2
4


Hub proteins: Cristae formation (magenta)
15 highest-kME members, most central at the top; black ring: FDR < 0.05
Training
Acute
TIMM13
MT-ATP6
GSTK1
ATP5MG
ACACA
SUCLG2
RBBP4
HSPA9
SLC25A11
TMEM11
PSAPL1
GCSH
RTN2
COQ3
AUH
HR
LR
HR - LR
HR
LR
HR - LR
logFC
(capped at ±1)
1.0
0.5
0.0
-0.5
-1.0
p < 0.05
p ≥ 0.05
-log10 p
1
2
4


Hub proteins: Response of EIF2AK4 GCN2 to amino acid def... (purple)
15 highest-kME members, most central at the top; black ring: FDR < 0.05
Training
Acute
RPLP0
RPL12
RPLP2
RPL7A
RPS16
RPL4
RPL19
VAPA
RPL13
RPL10A
PA2G4
RPL23A
RPL18
RPL30
RPL5
HR
LR
HR - LR
HR
LR
HR - LR
logFC
(capped at ±1)
1.0
0.5
0.0
-0.5
-1.0
p < 0.05
p ≥ 0.05
-log10 p
1
2
4